

[AEROTECH AVIATION SYSTEMS — COMPANY LOGO PLACEHOLDER]

AIRCRAFT ENGINE MAINTENANCE & INSPECTION MANUAL

Aircraft Model: AeroTech AT-320 | Engine: ATX-200 Turbofan

FICTIONAL TRAINING DOCUMENT

This is a synthetic, entirely fictional document created for AI/RAG research and educational demonstration purposes only. AeroTech Aviation Systems, the AT-320 aircraft, and the ATX-200 engine are fictional entities. This manual is not affiliated with, derived from, or representative of any real aircraft manufacturer, engine manufacturer, or airworthiness authority, and must never be used for actual aircraft maintenance, training, or certification.

Field	Value	Notes
Document Title	Aircraft Engine Maintenance & Inspection Manual	Training & demonstration use only
Revision	Revision 1.0	Initial synthetic release
Classification	Training & Demonstration Manual	Not for operational use
Aircraft Model	AeroTech AT-320 (fictional)	Narrow-body twin-engine, fictional
Engine Model	ATX-200 Turbofan (fictional)	High-bypass turbofan, fictional
Intended Use	RAG knowledge base / AI demo	Pairs with structured sensor + log data

Document Control

Revision History

Revision	Date	Author	Description of Change
0.1	2026-01-14	AeroTech Tech Pubs	Initial draft structure created
0.5	2026-02-03	AeroTech Tech Pubs	Added operating parameter tables and sensor chapter
1.0	2026-02-27	AeroTech Tech Pubs	First complete release for training/demo distribution

Document Control Table

Item	Detail
Document Owner	AeroTech Technical Publications Group
Review Cycle	Annual or upon major revision
Distribution	Internal training and AI/RAG demonstration use only
Approval Status	Approved for synthetic/demo use
Related Documents	AT-320 Structural Reference (fictional), ATX-200 Component Catalogue (fictional)

List of Figures

- Figure 2.1 — ATX-200 Turbofan Engine, Schematic Cross-Section
- Figure 2.2 — Compressor Stage Arrangement
- Figure 2.3 — Combustion Chamber Airflow Path
- Figure 2.4 — Turbine Section and Gearbox Interface
- Figure 7.1 — Troubleshooting Decision Tree: High Vibration Event
- Figure 7.2 — Troubleshooting Decision Tree: Low Oil Pressure Event
- Figure 7.3 — Troubleshooting Decision Tree: High Exhaust Gas Temperature Event

List of Tables

- Table 3.1 — Normal Operating Parameters Summary
- Table 4.1 — Primary Sensor Reference
- Table 5.1 — Failure Mode Reference Summary
- Table 6.1 — Scheduled Inspection Intervals
- Table 8.1 — Maintenance Decision Matrix

Table of Contents

Chapter 1 — Introduction
Chapter 2 — Engine Overview
Chapter 3 — Normal Operating Parameters
Chapter 4 — Aircraft Sensors
Chapter 5 — Failure Modes
Chapter 6 — Maintenance Procedures
Chapter 7 — Troubleshooting Guide
Chapter 8 — Maintenance Decision Matrix
Chapter 9 — Maintenance Work Orders
Chapter 10 — Appendix: Glossary & Codes

Chapter 1 — Introduction

1.1 Purpose

This manual establishes standardized inspection, monitoring, and maintenance procedures for the ATX-200 turbofan engine as installed on the AeroTech AT-320 airframe. It is intended to give maintenance personnel and engineering teams a consistent reference for normal operating ranges, sensor behaviour, common failure modes, and corrective actions.

NOTE

This is a fictional reference manual created for software demonstration purposes. All figures, thresholds, and procedures are illustrative and must not be applied to real aircraft.

1.2 Scope

This document covers line maintenance, scheduled inspection intervals, and first-level troubleshooting for the ATX-200 engine family. It does not cover heavy overhaul procedures, structural repairs to the airframe, or avionics systems unrelated to engine monitoring.

1.3 Responsibilities

Role	Responsibility
Line Maintenance Engineer	Performs scheduled inspections, records sensor readings, raises work orders for anomalies
Maintenance Supervisor	Reviews inspection reports, authorizes return-to-service, escalates critical findings
Reliability Engineer	Analyses recurring failure trends and updates inspection intervals
Quality Assurance	Audits compliance with inspection schedules and documentation standards

1.4 Safety Philosophy

Maintenance actions described in this manual follow a layered safety approach: continuous parameter monitoring, scheduled preventive inspection, and decision-based escalation when readings exceed defined thresholds. No corrective action should be deferred when a parameter is in the Critical range as defined in Chapter 3.

WARNING

Engines must never be returned to service when any monitored parameter remains in the Critical range without engineering sign-off, as defined in Chapter 8.

1.5 Maintenance Categories

- Line Maintenance — routine checks performed between flights
- Scheduled Maintenance — interval-based inspections (cycles, flight hours)
- Condition-Based Maintenance — triggered by sensor anomalies or exceedances

- **Unscheduled Maintenance** — triggered by pilot report or in-flight event

1.6 Aircraft Overview

The AeroTech AT-320 is a fictional narrow-body, twin-engine aircraft configured for short- to medium-haul commercial operation. Each aircraft is fitted with two ATX-200 high-bypass turbofan engines, one mounted under each wing.

Chapter 2 — Engine Overview

2.1 Turbofan Architecture

The ATX-200 is a dual-spool, high-bypass turbofan consisting of a fan, low- and high-pressure compressors, an annular combustion chamber, high- and low-pressure turbines, and an accessory gearbox. Bypass air provides the majority of thrust, while core airflow drives the compressor-turbine spool.

Figure 2.1 — ATX-200 Turbofan Engine, Schematic Cross-Section. [Vector diagram placeholder: fan inlet, low-pressure compressor, high-pressure compressor, combustion chamber, high-pressure turbine, low-pressure turbine, exhaust nozzle, labelled left to right.]

2.2 Compressor

The compressor section consists of a low-pressure (LP) compressor and a high-pressure (HP) compressor arranged in series. Each stage progressively increases air pressure prior to combustion.

Figure 2.2 — Compressor Stage Arrangement. [Vector diagram placeholder: rotor and stator blade rows, LP and HP sections shaded differently.]

2.3 Combustion Chamber

An annular combustion chamber mixes compressed air with fuel for ignition. Even fuel-air distribution is critical to preventing localized hot spots that accelerate liner wear.

Figure 2.3 — Combustion Chamber Airflow Path. [Vector diagram placeholder: fuel nozzles, swirler, primary and dilution air zones.]

2.4 Turbine

The high-pressure turbine extracts energy to drive the HP compressor; the low-pressure turbine drives the fan and LP compressor. Turbine blades operate at the highest temperatures in the engine and are the primary subject of borescope inspection.

Figure 2.4 — Turbine Section and Gearbox Interface. [Vector diagram placeholder: HP turbine, LP turbine, shaft coupling, accessory gearbox.]

2.5 Gearbox

The accessory gearbox is driven off the HP spool and powers the fuel pump, oil pump, hydraulic pump, and electrical generator.

2.6 Fuel System

Fuel is metered by an electronic engine control unit based on throttle position, altitude, and temperature inputs, then delivered to the combustion chamber through a ring of fuel nozzles.

2.7 Lubrication System

A closed-loop oil system lubricates and cools the main bearings and gearbox. Oil temperature and pressure are continuously monitored as primary indicators of bearing and gearbox health.

2.8 Hydraulic System

Engine-driven hydraulic pumps supply pressure for actuation of variable stator vanes and bleed valves.

2.9 Cooling System

Bleed air from the compressor is routed internally to cool turbine blades and discs, and externally to manage nacelle temperatures.

Chapter 3 — Normal Operating Parameters

Table 3.1 lists the primary monitored parameters for the ATX-200 engine, together with their normal, warning, and critical operating bands and the corresponding recommended action. These bands form the basis for all condition-based maintenance logic referenced throughout this manual.

Parameter	Normal	Warning	Critical	Recommended Action
Engine Temperature (°C)	640 – 676	676 – 696	> 696	Inspect cooling air ducts and combustion liner
Exhaust Gas Temp. (°C)	615 – 648	648 – 668	> 668	Inspect turbine section; review fuel scheduling
Oil Temperature (°C)	82 – 92	92 – 95	> 95	Inspect oil cooler and scavenge lines
Oil Pressure (PSI)	55 – 61	52 – 55	< 52	Inspect oil pump, filter, and seals
Fuel Flow (kg/hr)	2235 – 2338	2338 – 2382	> 2382	Check fuel metering unit calibration
Compressor Pressure (PSI)	38.5 – 41.5	41.5 – 42.7	> 42.7	Inspect variable stator vane schedule
Engine Vibration (mm/s)	0 – 3.1	3.1 – 4.7	> 4.7	Inspect bearing assembly and rotating components
Hydraulic Pressure (PSI)	2950 – 3050	—	—	Stable parameter; verify against reservoir level only
Engine RPM (N1, RPM)	9700 – 9900	—	—	Stable parameter; cross-check governor calibration
Ambient Humidity (%RH)	20 – 88	88 – 95	> 95	Monitor for ice/water ingestion risk
Ambient Temperature (°C)	-10 – 35	35 – 40	> 40	Apply hot-weather start procedure
Outside Air Temp. (°C)	-19 – 28	28 – 37	> 37	Review derate tables before takeoff

NOTE

Warning range readings require increased monitoring frequency and entry into the condition-based inspection cycle. Critical range readings require immediate corrective action per Chapter 8 before further flight.

NOTE

Hydraulic Pressure and Engine RPM are continuously monitored but operate within a narrow, stable band under normal conditions across the fleet. These parameters are not primary indicators of impending failure in isolation and should be assessed alongside vibration, temperature, and pressure trends rather than as standalone triggers.

Chapter 4 — Aircraft Sensors

Table 4.1 summarizes the primary sensors used to monitor ATX-200 engine health.

4.1 Engine Temperature Sensor

Attribute	Detail
Purpose	Monitors core engine case temperature as a primary thermal health indicator
Units	°C
Location	Compressor case, mid-span thermocouple
Inspection Method	Signal trend review each flight cycle; calibration check at 250-hour interval
Expected Behaviour	Stable within 640–676°C across normal power range, rising predictably with thrust setting
Failure Symptoms	Sustained reading above 676°C, erratic fluctuation, or drift from EGT-correlated trend

4.2 Vibration Sensor (Accelerometer)

Attribute	Detail
Purpose	Detects rotor imbalance, bearing wear, and rotating assembly degradation
Units	mm/s (RMS)
Location	Mounted on forward and aft engine casing
Inspection Method	Visual mount check; signal trend review each flight cycle
Expected Behaviour	Stable within 0–3.1 mm/s under normal conditions; gradual upward trend signals developing wear
Failure Symptoms	Sustained reading above 3.1 mm/s, erratic spikes, or signal dropout

4.3 Exhaust Gas Temperature (EGT) Probe

Attribute	Detail
Purpose	Monitors turbine inlet/exit gas temperature as a primary indicator of hot-section health
Units	°C
Location	Turbine exhaust duct, multiple probes averaged
Inspection Method	Borescope check of probe tip during scheduled inspection
Expected Behaviour	Stable within 615–648°C at steady throttle; rises predictably with EGT-to-temperature ratio maintained
Failure Symptoms	Sustained reading above 648°C, probe open/short circuit, frozen reading, or

Attribute	Detail
	abnormal spread between probes

4.4 Oil Temperature Sensor

Attribute	Detail
Purpose	Monitors lubrication system thermal condition
Units	°C
Location	Oil return line, downstream of bearing chambers
Inspection Method	Calibration check at scheduled intervals
Expected Behaviour	Stable within 82–92°C; tracks engine temperature with minor lag
Failure Symptoms	Sustained reading above 92°C, particularly when paired with a falling oil pressure trend

4.5 Oil Pressure Transducer

Attribute	Detail
Purpose	Monitors lubrication system pressure and bearing/gearbox feed adequacy
Units	PSI
Location	Main oil supply line, downstream of oil pump
Inspection Method	Calibration check at scheduled intervals
Expected Behaviour	Stable within 55–61 PSI, proportional to RPM
Failure Symptoms	Sustained reading below 55 PSI, fluctuating signal, or zero reading

4.6 Fuel Flow Transmitter

Attribute	Detail
Purpose	Measures fuel mass flow to the combustion chamber
Units	kg/hr
Location	Fuel supply line upstream of fuel manifold
Inspection Method	Flow calibration check during scheduled inspection
Expected Behaviour	Stable within 2235–2338 kg/hr at steady throttle; tracks throttle position with minimal lag
Failure Symptoms	Sustained reading above 2338 kg/hr at constant throttle, or mismatch with throttle position

4.7 Compressor Discharge Pressure Sensor

Attribute	Detail
Purpose	Monitors HP compressor discharge pressure
Units	PSI
Location	HP compressor exit duct
Inspection Method	Leak check of pressure tap during inspection
Expected Behaviour	Stable within 38.5–41.5 PSI, rising smoothly with RPM
Failure Symptoms	Sustained reading above 41.5 PSI, pressure plateau, or oscillation indicating stall risk

4.8 Hydraulic Pressure Transducer

Attribute	Detail
Purpose	Monitors engine-driven hydraulic pump output
Units	PSI
Location	Hydraulic pump discharge line
Inspection Method	Reservoir level check and pressure verification at scheduled inspection
Expected Behaviour	Stable within 2950–3050 PSI across all normal operating conditions
Failure Symptoms	Reading outside the stable band, combined with visible fluid loss or reservoir depletion

4.9 Engine Speed Sensor (N1 Tachometer)

Attribute	Detail
Purpose	Monitors low-pressure spool rotational speed
Units	RPM
Location	LP shaft, forward of fan hub
Inspection Method	Calibration check against governor reference at scheduled inspection
Expected Behaviour	Stable within 9700–9900 RPM at commanded power settings
Failure Symptoms	Reading outside the stable band without corresponding throttle command, or erratic governor response

4.10 Ambient Condition Sensors (Temperature, Humidity, OAT)

Attribute	Detail
Purpose	Provide environmental context for engine performance correction and icing risk assessment

Attribute	Detail
Units	°C / %RH
Location	Aircraft skin probes and pitot-static system
Inspection Method	Cross-check against ground station reference during pre-flight inspection
Expected Behaviour	Tracks local environmental conditions; used to contextualize engine parameter trends rather than as a standalone health indicator
Failure Symptoms	Reading inconsistent with ground reference, or humidity above 88% combined with OAT near freezing, indicating icing risk

Chapter 5 — Failure Modes

This chapter describes the failure modes most frequently identified through condition monitoring on the ATX-200 fleet, together with their symptoms, likely causes, detection logic, severity, and recommended response. Sections 5.1 through 5.4 represent the primary failure modes tracked by the fleet monitoring programme; Sections 5.5 through 5.10 cover additional failure modes relevant to comprehensive engine maintenance.

5.1 Primary Fleet-Monitored Failure Modes

5.1.1 Bearing Wear

Attribute	Detail
Symptoms	Rising vibration trend, audible roughness at idle, gradual engine and EGT temperature creep
Possible Causes	Lubrication breakdown, contamination, fatigue of rolling elements
Detection Logic	Vibration sustained above 3.1 mm/s, frequently accompanied by engine temperature above 696°C and EGT above 668°C
Severity	Medium to High
Immediate Action	Schedule borescope and oil debris check within 20 flight cycles
Engineering Recommendation	Inspect and replace bearing per Chapter 6.5 if vibration remains in the Critical band after re-check

5.1.2 Sensor Fault

Attribute	Detail
Symptoms	Erratic, frozen, or inconsistent reading on one or more channels without a corresponding physical trend
Possible Causes	Wiring fault, connector corrosion, sensor element drift or failure
Detection Logic	Reading inconsistent with cross-referenced parameters; vibration and temperature readings elevated but below Critical thresholds with no consistent physical pattern
Severity	Low to Medium
Immediate Action	Cross-check with redundant sensor or ground test set where available
Engineering Recommendation	Calibrate or replace the affected sensor per Chapter 9 work order procedure

5.1.3 Hydraulic Leak

Attribute	Detail
Symptoms	Hydraulic pressure outside the stable 2950–3050 PSI band, visible fluid loss, elevated vibration

Attribute	Detail
Possible Causes	Seal wear, fitting failure, pump wear
Detection Logic	Hydraulic pressure deviation combined with vibration above 4.7 mm/s and oil pressure trending below 55 PSI
Severity	Medium to High
Immediate Action	Inspect reservoir level and visible lines before further flight
Engineering Recommendation	Replace affected seal or fitting; verify pump output meets reference band

5.1.4 Oil Leakage

Attribute	Detail
Symptoms	Oil pressure decline, visible staining, low oil quantity, elevated vibration
Possible Causes	Seal degradation, loose fitting, cracked supply or return line
Detection Logic	Oil pressure below 52 PSI combined with vibration above 4.7 mm/s
Severity	High
Immediate Action	Ground aircraft; perform visual leak check before further flight
Engineering Recommendation	Replace seal or line; verify root cause and oil quantity before return to service

5.2 Additional Failure Modes

5.2.1 Fuel Leakage

Attribute	Detail
Symptoms	Fuel odor, visible wetting, unexplained flow discrepancy
Possible Causes	Fitting failure, line chafing, seal wear
Detection Logic	Cross-check fuel flow transmitter against fuel quantity indication
Severity	High
Immediate Action	Ground aircraft immediately; isolate fuel supply
Engineering Recommendation	Inspect and replace affected fuel line or fitting

5.2.2 Compressor Stall

Attribute	Detail
Symptoms	Bang/surge noise, RPM fluctuation, EGT spike

Attribute	Detail
Possible Causes	Foreign object damage, variable vane schedule fault, icing
Detection Logic	Compressor discharge pressure above 42.7 PSI combined with EGT spike
Severity	High
Immediate Action	Reduce power; inspect for foreign object damage
Engineering Recommendation	Inspect variable stator vane actuation; borescope compressor

5.2.3 Turbine Blade Damage

Attribute	Detail
Symptoms	Localized EGT spike, vibration increase
Possible Causes	Foreign object ingestion, thermal fatigue, erosion
Detection Logic	Borescope inspection findings combined with EGT asymmetry above 668°C
Severity	High
Immediate Action	Ground aircraft; full borescope inspection
Engineering Recommendation	Blade replacement or repair per engineering disposition

5.2.4 Combustion Instability

Attribute	Detail
Symptoms	Audible rumble, EGT fluctuation at steady throttle
Possible Causes	Fuel nozzle fouling, uneven fuel distribution
Detection Logic	EGT spread across probes exceeds expected tolerance at constant fuel flow
Severity	Medium
Immediate Action	Inspect fuel nozzles for fouling/damage
Engineering Recommendation	Clean or replace fuel nozzles; verify fuel pressure

5.2.5 High Exhaust Gas Temperature

Attribute	Detail
Symptoms	EGT consistently elevated for a given power setting
Possible Causes	Turbine wear, cooling air blockage, fuel scheduling fault
Detection Logic	EGT trend exceeds 668°C at constant throttle position

Attribute	Detail
Severity	Medium to High
Immediate Action	Reduce thrust; schedule inspection
Engineering Recommendation	Inspect turbine cooling passages and fuel control unit

5.2.6 Low Oil Pressure (Isolated)

Attribute	Detail
Symptoms	Oil pressure below normal range without accompanying vibration increase
Possible Causes	Pump wear, blocked filter, low oil quantity
Detection Logic	Oil pressure transducer reading below 55 PSI in isolation
Severity	Medium to High
Immediate Action	Check oil quantity and filter before further flight
Engineering Recommendation	Replace oil pump or filter as indicated by inspection

Chapter 6 — Maintenance Procedures

6.1 Scheduled Inspection Intervals

Inspection	Trigger	Key Checks
Post-Flight Walkaround	Every landing	Visual leak check, fluid stains, foreign object damage
50-Cycle Inspection	Every 50 flight cycles	Oil quantity, vibration trend review, visual fan blade check
100-Cycle Inspection	Every 100 flight cycles	Borescope of hot section, fuel nozzle visual check
250-Hour Inspection	Every 250 flight hours	Full sensor calibration check, hydraulic system check
Post-Vibration-Event Inspection	After abnormal vibration alert	Bearing inspection, oil debris analysis
Post-Exceedance Inspection	After temperature/pressure exceedance	Borescope of affected section, engineering review

6.2 Oil System Inspection Procedure

1. Confirm engine has cooled to ambient-safe handling temperature.
2. Check oil quantity against minimum dispatch level.
3. Inspect oil filter for debris; record any metallic content found.
4. Verify oil pressure transducer reading against ground-test reference.
5. Record results on Inspection Report (see Chapter 9).

CAUTION

Allow adequate cooling time before opening oil system access panels to avoid burn hazard.

6.3 Hydraulic System Inspection Procedure

1. Inspect reservoir fluid level against minimum mark.
2. Check visible hydraulic lines for seepage or chafing.
3. Verify pump discharge pressure against reference table in Chapter 3.

6.4 Compressor Inspection Procedure

1. Perform visual inspection of fan and LP compressor blades for nicks or erosion.
2. Borescope HP compressor stages per scheduled interval.
3. Cross-check compressor discharge pressure sensor against Chapter 3 reference ranges.

6.5 Bearing Inspection Procedure

1. Review vibration sensor trend data for the preceding 10 flight cycles.
2. Perform oil debris analysis for metallic contamination.
3. Borescope accessible bearing housings where vibration trend exceeds 3.1 mm/s.
4. Replace bearing where vibration trend exceeds 4.7 mm/s or oil debris confirms wear.
5. Disposition per Maintenance Decision Matrix (Chapter 8).

Chapter 7 — Troubleshooting Guide

This chapter presents decision-tree style troubleshooting logic for common exceedance events. Each tree should be read top to bottom.

7.1 Figure 7.1 — High Vibration Event

Step 1: Vibration reading exceeds Warning threshold (3.1 mm/s)

Step 2: Check oil pressure and oil debris content

Step 3a: If oil pressure normal (above 55 PSI) and no debris — continue monitoring, re-check next cycle

Step 3b: If oil pressure below 55 PSI or debris present — proceed to bearing inspection

Step 4: If bearing inspection confirms wear — replace bearing per Chapter 6.5

Step 5: If vibration exceeds Critical threshold (4.7 mm/s) at any point — schedule inspection within 20 flight cycles per Chapter 8

7.2 Figure 7.2 — Low Oil Pressure Event

Step 1: Oil pressure reading falls below Warning threshold (55 PSI)

Step 2: Check oil quantity against minimum dispatch level

Step 3a: If oil quantity is low — replenish and monitor; investigate consumption rate

Step 3b: If oil quantity is normal — inspect oil filter and pump

Step 4: If pressure falls below Critical threshold (52 PSI) combined with elevated vibration — assess for oil leakage or hydraulic leak per Chapter 5.1

7.3 Figure 7.3 — High Exhaust Gas Temperature Event

Step 1: EGT reading exceeds Warning threshold (648°C) at steady throttle

Step 2: Cross-check EGT spread across all probes and corresponding engine temperature reading

Step 3a: If spread is within tolerance and engine temperature is below 696°C — likely gradual turbine wear; schedule borescope

Step 3b: If spread exceeds tolerance — suspect fuel nozzle fouling or localized hot spot

Step 4: If EGT exceeds Critical threshold (668°C) — reduce power and inspect prior to further flight

Chapter 8 — Maintenance Decision Matrix

Table 8.1 provides the risk-based framework used to translate a computed engine risk score into an aircraft maintenance status and corresponding recommended action. The risk score is derived from a weighted combination of vibration trend, temperature margin, oil pressure margin, and time since last overhaul, expressed as a percentage.

Risk Score	Status	Typical Indicators	Recommended Action
0 – 35%	Healthy	All parameters within Normal band; vibration below 3.1 mm/s	No action required; continue scheduled monitoring
35 – 60%	Monitor	One or more parameters in Warning band; vibration 3.1–4.7 mm/s	Increase monitoring frequency; schedule inspection within defined interval
60 – 100%	Maintenance Due	One or more parameters in Critical band; vibration above 4.7 mm/s	Inspect or replace affected component within 20 flight cycles

NOTE

Risk score thresholds are calibrated against fleet-wide condition monitoring data and are reviewed periodically by the Reliability Engineering group. A Maintenance Due status does not by itself require grounding the aircraft; disposition is determined by the specific parameter in the Critical band and the corresponding failure mode guidance in Chapter 5.

Chapter 9 — Maintenance Work Orders

This chapter provides templates used to document inspection and maintenance activity.

9.1 Inspection Report Template

Field	Entry
Aircraft Registration / ID	_____
Engine Serial Number	_____
Flight Cycles / Flight Hours	_____
Cycles Since Last Overhaul	_____
Inspection Type	_____
Date / Time	_____
Engine Temperature (°C)	_____
Exhaust Gas Temperature (°C)	_____
Oil Temperature / Pressure (°C / PSI)	_____
Engine Vibration (mm/s)	_____
Compressor Pressure (PSI)	_____
Fuel Flow (kg/hr)	_____
Hydraulic Pressure (PSI) / Engine Speed (RPM)	_____
Computed Risk Score (%)	_____
Findings / Detected Failure Mode	_____
Inspector Name	_____

9.2 Maintenance Completion Form

Field	Entry
Work Order Number	_____
Task Description	_____
Parts Replaced	_____
Test Results	_____
Return-to-Service Status	_____

9.3 Replacement Checklist

- Confirm correct part number and serial traceability
- Verify torque values per component specification
- Perform post-installation functional test
- Update component life/cycle tracking record

9.4 Root Cause Analysis Template

Field	Entry
Problem Statement	
Contributing Factors	
Root Cause	
Corrective Action	
Preventive Action	

9.5 Sign-Off

Role	Name	Signature	Date
Engineer Sign-off			
Supervisor Sign-off			

Chapter 10 — Appendix

10.1 Engineering Glossary

Term	Definition
EGT	Exhaust Gas Temperature — temperature of gases exiting the turbine section
N1	Rotational speed of the low-pressure spool, expressed as a percentage
Bleed Air	Compressed air extracted from the compressor for cooling or pneumatic use
Borescope	Optical inspection tool used to examine internal engine components
Disposition	Engineering decision on repair, replace, or continue-monitoring status

10.2 Abbreviations

Abbreviation	Meaning
HP	High Pressure
LP	Low Pressure
PSI	Pounds per Square Inch
RPM	Revolutions Per Minute
RH	Relative Humidity

10.3 Inspection Symbols & Maintenance Codes

Code	Meaning
INSP-A	Routine post-flight inspection
INSP-B	Cycle-based scheduled inspection
INSP-C	Condition-based inspection (triggered by exceedance)
WO-R	Work order — repair
WO-X	Work order — replacement

10.4 Revision Notes

This Revision 1.0 release represents the first complete synthetic version of this training manual, intended solely for use as reference content in AI/RAG demonstration systems.